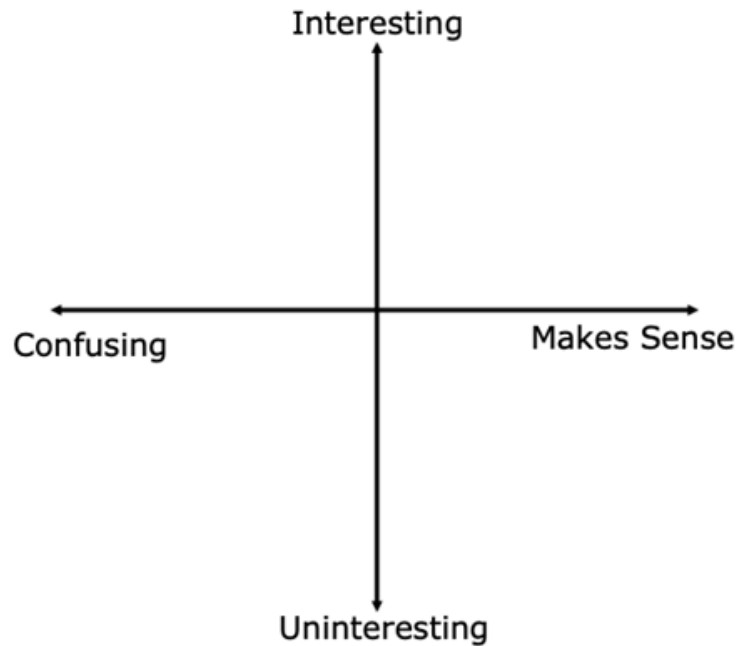




8.3.6 Wrap-Up: Reflection

1. Plot a point on the grid to indicate your level of interest and how much sense you made of today's learning targets.



2. Explain why you placed the point where you did.

Unit 8, Lesson 4: Kites and Parallelograms



Benchmark

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.



Learning Target

- ☐ I can use properties of parallelograms, kites, and special parallelograms to solve mathematical and real-world problems.



8.4.1 Warm-Up: Canstruction

Each year just before Thanksgiving there are competitions held all over the world to build unique Canstructions with canned food that is collected from the community.



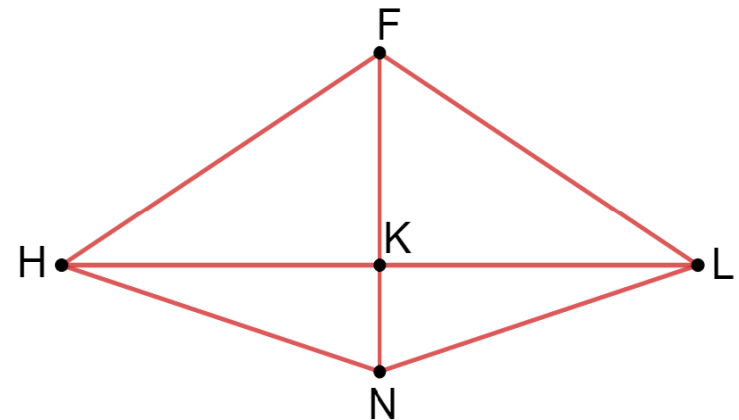
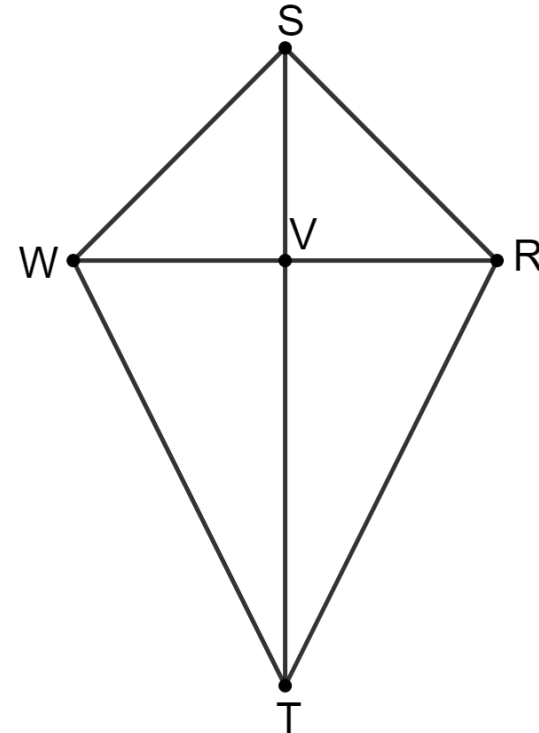
The food is then given to local food banks. The picture shows the Canstruction called "Downside Up," which was the winning exhibit from Canstruction 2010 in New York City.

1. What do you wonder about "Downside Up"?
2. Make an educated guess at the number of cans used to build "Downside Up". Show your work.
3. Compare your educated guess with your partner's. Describe how close your guess was to your partner's guess.



8.4.2 Exploration: What Is a Kite?

Kite $SRTW$ with diagonals $\overline{ST}$ and $\overline{WR}$ intersecting at point V and $FLNH$ with diagonals $\overline{FN}$ and $\overline{HL}$ intersecting at point K are shown.



- Using a ruler, work with a partner to fill in the chart below with the appropriate measurements. Each partner should find the measurements for one kite and share their results with the other.

<i>SRTW</i>		<i>FLNH</i>	
Segment	Measurement	Segment	Measurement
$\overline{SR}$		$\overline{FL}$	
$\overline{SW}$		$\overline{FH}$	
$\overline{RT}$		$\overline{HN}$	
$\overline{WT}$		$\overline{LN}$	
$\overline{ST}$		$\overline{FN}$	
$\overline{SV}$		$\overline{FK}$	
$\overline{TV}$		$\overline{NK}$	
$\overline{WR}$		$\overline{HL}$	
$\overline{WV}$		$\overline{HK}$	
$\overline{RV}$		$\overline{LK}$	

- What do you notice about $\overline{SR}$ and $\overline{SW}$ in *SRTW* and $\overline{FL}$ and $\overline{FH}$ in *FLNH*?
- What do you notice about $\overline{RT}$ and $\overline{WT}$ in *SRTW* and $\overline{HN}$ and $\overline{LN}$ in *FLNH*?

- These segment pairs are **adjacent** sides of a kite.
 - Make a conjecture about the measures of adjacent sides of a kite.
 - Make a conjecture about the lengths of diagonals of a kite.
- Using a protractor, work with your partner to fill in the chart below with the appropriate measurements. Each partner should find the measurements for one kite and share their results with the other.

<i>SRXW</i>		<i>FLNH</i>	
Angle	Measurement	Angle	Measurement
$\angle SVR$		$\angle FKL$	
$\angle WSR$		$\angle HFL$	
$\angle RTW$		$\angle LNH$	
$\angle TWS$		$\angle FHN$	
$\angle TRS$		$\angle FLN$	

- Make a conjecture about what happens when the diagonals of a kite intersect.
- Make a conjecture about the measures of opposite angles of a kite.

A **kite** is a quadrilateral with exactly two pairs of adjacent congruent sides.



A **kite** has the following properties:

- One pair of opposite congruent angles
- Perpendicular diagonals
- One bisected diagonal



8.4.3 Collaboration: Quadrilaterals

- With your partner, use the bank of phrases to fill in the chart comparing quadrilaterals. Statements can be used more than once.

A. Opposite sides are congruent.	B. Opposite sides are parallel.
C. All sides are congruent.	D. Diagonals bisect each other.
E. Opposite angles are congruent.	F. Diagonals are congruent.
G. All angles are congruent.	H. Diagonals are perpendicular.
I. Two pairs of adjacent sides are congruent.	J. One diagonal is bisected.
K. One pair of opposite angles are congruent.	L. Opposite angles are supplementary.
M. One pair of parallel sides	N. Non-parallel sides are congruent.
O. Two pairs of congruent angles	

- Write the letter(s) from the bank.

Quadrilateral	Sides	Angles	Diagonals
Parallelogram			
Rectangle			
Square			
Rhombus			
Kite			

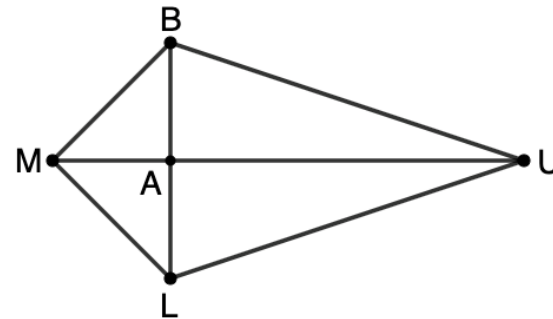
- b. Compare your chart with that of another pair. Add in anything you may have missed.

- c. Discuss with your partner whether or not kites can be classified as parallelograms.

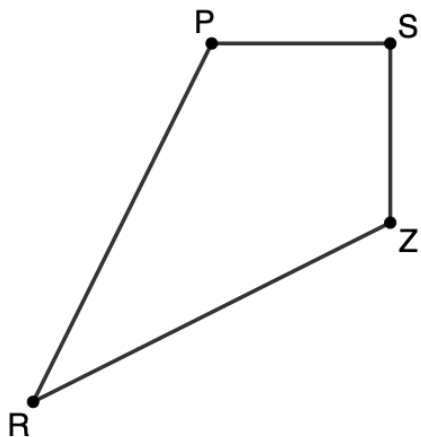


8.4.4 Guided Instruction: Quadrilaterals and Algebra

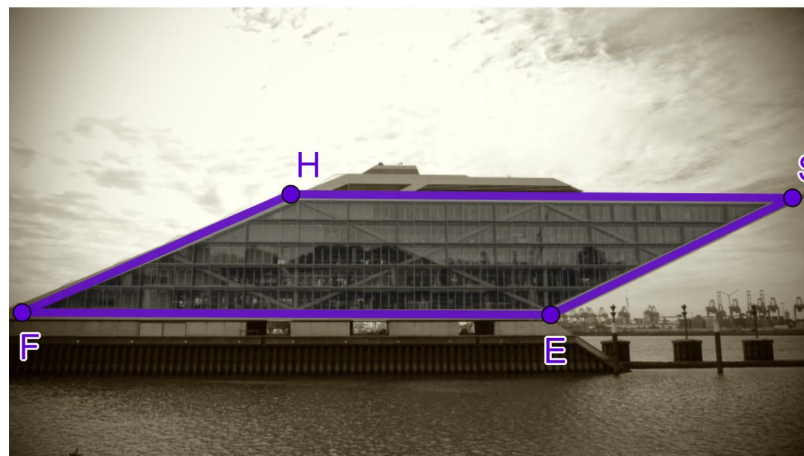
1. $MBUL$ with diagonals $\overline{BL}$ and $\overline{MU}$ intersecting at point A is shown. Find the value of x that makes $MBUL$ a kite if $BA = 3x - 1$, $LA = 17$, $MA = 2x$, and $UA = 23$.



2. Find the values of x and y that make $PSZR$ a kite if $PS = 5x - 4$, $PR = 8y - 4$, $SZ = 3y + 1$, and $RZ = 36$.



3. The Dockland office building is one of the most famous buildings in Hamburg, Germany. Built in 2005, this building houses roughly 9,000 square meters of office space and breathtaking views of the Elbe River, but the size and the views are not what makes this building so famous. The building itself is in the shape of a parallelogram, $HSEF$.

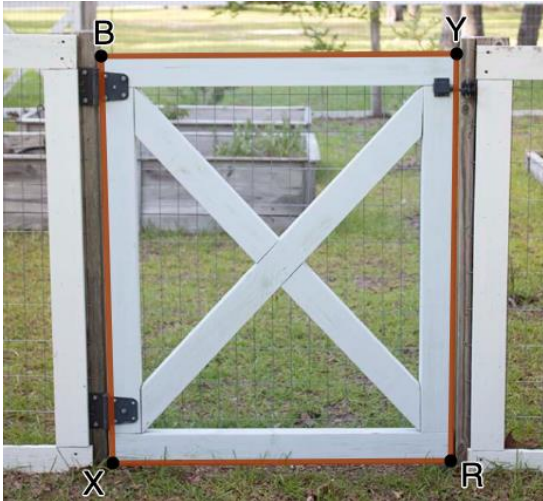


The perimeter of the Dockland building measures approximately 238 meters. If the length of $\overline{FE}$ is 2.8 times the length of $\overline{FH}$, what are the measures of $\overline{HS}$, $\overline{SE}$, $\overline{FE}$, and $\overline{FH}$? Round your answers to the nearest thousandth of a meter.



8.4.5 Your Turn!

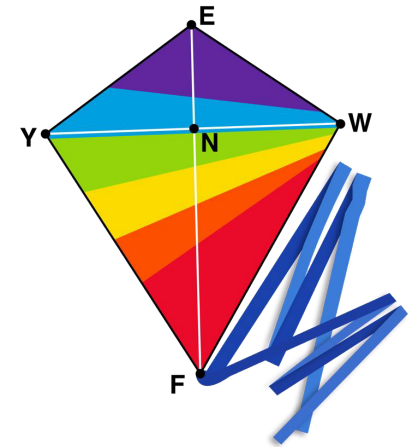
1. Oliver's carpentry company, Wood 4 You, has been hired to build a fence around a vegetable garden. The gate must be a perfect rectangle. The gate Oliver built is shown, where quadrilateral $BYRX$ outlines the gate.



Oliver wants to be absolutely sure that the gate is a rectangle. Describe two different ways Oliver can use his tape measure to ensure the gate is a true rectangle.

2. Piper is entering a kite-making contest. The kite she designed is shown, where the kite has been labeled as $YEFW$ with measurements $EW = 7.5$, $YN = 6$, and $FN = 10.2$, given in inches.

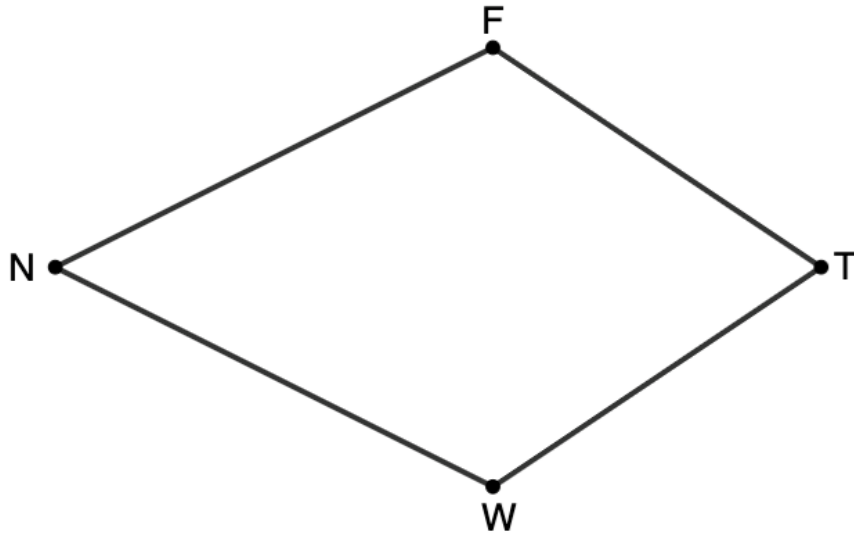
Piper wishes to add a strip of glitter fabric along diagonals $\overline{EF}$ and $\overline{YW}$ that will overlap at their intersection point N , as well as along the perimeter of the kite. How many inches of glitter fabric are needed to cover the diagonals and the perimeter of the kite?





8.4.6 Wrap-Up: Cool Down

1. Kite $FTWN$ is shown. Find the value of x if $m\angle N = 53^\circ$, $m\angle T = 67^\circ$, and $m\angle F = \left(\frac{1}{2}x - 8\right)^\circ$.



Unit 8, Lesson 5: Trapezoids



Benchmarks

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can prove relationships and theorems about trapezoids.
- ☐ I can use relationships and theorems about trapezoids.